2022 4th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE) 29-31 December 2022, Rajshahi-6204, Bangladesh

Segmented-Sparse-PCA for Hyperspectral Image Classification

Md. Shahriar Haque Shemul
Department of CSE
HSTU, Dinajpur - 5200, Bangladesh
shahriarhaque254@gmail.com

Md Moshiur Rahman
Department of CSE
HSTU, Dinajpur - 5200, Bangladesh
rahmanmoshiur623@gmail.com

Shabbir Ahmed
Department of CSE
HSTU, Dinajpur - 5200, Bangladesh
shabbirtkg@gmail.com

Md. Abu Marjan
Department of CSE
HSTU, Dinajpur - 5200, Bangladesh
marjan@hstu.ac.bd

Md. Palash Uddin
Department of CSE
HSTU, Dinajpur - 5200, Bangladesh
palash_cse@hstu.ac.bd

Masud Ibn Afjal
Department of CSE
HSTU, Dinajpur - 5200, Bangladesh
masud@hstu.ac.bd

***Abstract*—Remote sensing hyperspectral images (HSI) are typically acquired to obtain critical information about the land covers via adjacent cramped spectral wavelength bands. The classification performance appears inadequate when every original HSI band is in use. In order to improve performance of classification, band (dimensionality) reduction approaches using feature extraction as well as feature selection techniques are widely utilized to mitigate this. Despite the widespread usage of Principal Component Analysis (PCA) in HSI feature reduction, it frequently struggles to assess the local beneficial properties of the HSI as it analyzes solely the HSI's global data. Segmented-PCA (SPCA) and Sparse-PCA are initiatives to replace the PCA. In this paper, we propose the Segmented-Sparse-PCA (SSPCA) feature extraction approach nto exploit the amenities of both SPCA and Sparse- PCA. In particular, we first segment the entire dataset into a number of strongly correlated spectral band subgroups and then apply Sparse-PCA to each subgroup separately. Afterward, the results are analyzed through experimenting over the mixed agricultural Indian Pines HSI dataset using a per-pixel Support Vector Machine (SVM) classifier. The experimental results exhibit that the overall classification performance of the proposed SSPCA (91.024%) performs better compared to every other investigated approaches: PCA (83.554%), SPCA (86.774%), Sparse-PCA (88.39%) and the entire original bands of HSI (82.997%).**

***Index Terms*—Hyperspectral image classification, Feature extraction, Remote sensing, PCA, Segmented PCA, Sparse PCA, Segmented Sparse-PCA**

I. Introduction

Hyperspectral imaging (HSI) is a heartening discipline that brings together the advantages of optical spectroscopy as an analytical tool and the two-dimensional object viewing offered by spectral imaging. It involves the simultaneous acquisition and integration of spectral and spatial properties in image space. As such, it offers high spectral resolutions via hundreds of small and continuous spectral wavelength bands, which can be beneficial in a variety of applications [1]. To acquire information about a given ground surface, the wavelength range for capturing HSIs is usually from 0.4 μm to 3.0 μm, encompassing the EM-wave spectrum between visible light and near-infrared [2]. The spectral resolution delineated by nm reflects HSI's narrow absorption features. HSI data typically comprises a hypercube with 2D spatial information and a tertiary dimension within a hypercube that offers spectral features regarding the captured objects. Therefore, the complete HSI data contains $X \times Y \times F$, with dimention X indicating rows, Y signifying columns, and F identifying spectral information.

Although adjoining bands exhibit information that is significantly correlated, not every HSI band may posses an equal deal of information [3]. When the whole original HSI is used for real-world use, the classification performance is occasionally insufficient. Forbye, for improved classification performance, an approximately exponential quantity of samples to the amount of features is necessary. Although practically, gathering an ample quantity of samples for training is difficult. Therefore, feature reduction techniques, which are separated into feature selection and feature reduction methods, are required to reduce the HSI bands for enhanced classification accuracy. The original data is frequently used in feature selection methods together with a variety of criteria for searching to seek out the pertinent bands [4]. These searching criteria are oftentimes challenging since the combinatorial exploration requires high computational costs [5]. On the contrary, feature extraction, a better alternative strategy for reducing dimensionality, obtains the most valuable insight about the HSI after executing several mathematical transformations on the dataset [6]. As a way of retrieving crucial and inherent attributes from the HSI, the unsupervised as well as supervised feature extraction approaches in turn deploy nonlinear and linear transformations. The training data, which can be computationally demanding, is critical to the success of supervised feature extraction approaches. Unsupervised feature extraction strategies, on the other hand, are used to acquire pertinent traits of the data devoid of prior knowledge [7]. Seamise-CNN, generalized embedding regression, mixture of factor analysis, and other supervised feature extraction procedures are also relied on for HSI classification [8] [9].

The widely prominent unsupervised feature extraction tool used in hyperspectral imagery is principal component analysis

2022 4th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE) | 979-8-3503-2054-1/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICECTE57896.2022.10114537

(PCA). It requires the data matrix to be centered and could not be capable of working with sparse datasets directly. In addition, PCA demands the data matrix to be centered and could not be capable of working with a sparse dataset directly while a more concise and understandable representation is produced by sparse-PCA, stressing specifically which of the original features leads to the fluctuations between instances. To what follows, Sparse-PCA substantially supersedes PCA in the HSI context. We introduce the SSPCA feature extraction strategy, which segments the HSI data down to a set of correlated band subgroupings and then performs Sparse-PCA separately towards extracting the whole dataset's global and local properties, for improving the classification performance. Moreover, the classification performance using proposed SSPCA is analogised to the results of using the original dataset, PCA, SPCA, and Sparse-PCA. To this end, Listed below are this paper's core contributions:

- Presenting the Segmented-Sparse-PCA (SSPCA) feature extraction by fusing the benefits of Sparse-PCA along with SPCA.
- A pragmatic delineation using a series of experiments for validating the superiority of the proposed SSPCA.

The following is how the rest of the writing is structured: Section II-A discusses the overall idea and derivation of the proposed SSPCA approach. Comparative result analysis and test settings are provided in Section III while Section IV summarizes the studies' results and insights.

II. Feature Extraction

A. Proposed SSPCA

If the HSI bands are substantially connected, the feature extraction techniques like standard PCA retrieves operative characteristics. PCA retrieves efficient features when the original bands are strongly correlated and have a large covariance matrix of size F×F, which needs a greater computing overhead [10]. Furthermore, correlations between contiguous HSI bands are frequently greater than correlations between bands further away, with high correlations emerging in blocks. Therefore, Segmentation is presented to improve the performance of the traditional PCA by eliminating weak correlation between strongly correlated blocks [10]. On the other hand,sparse principal components result in a more concise, comprehensible depiction. The aim of the Sparse-PCA, a PCA modification, is to extract the sparse components that effectively reassemble the data. And, it has previously been used for HSIs [11]. Whereas, The drawback of PCA is that the components extracted have solely dense expressions, and have non-zero coefficients when presented as linear combinations of the original data. This makes explication more complicated. Often , the true basal components can be envisioned more intuitively as sparse vectors. Thus segmented data is fitted into the Sparse-PCA ($O(p^3)$) [12] in this work.

The dataset is initially broken down into L segments in our proposed SSPCA based on the strong correlation among the bands. Let $\mathbf{D}_i$ stand for the dataset $\mathbf{D}$'s i^{th} band segment,

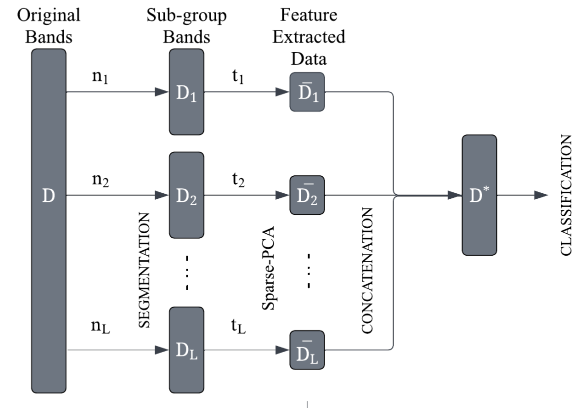

Fig. 1. Conceptual layout of the proposed SSPCA.

TABLE I
THE INDIAN PINES HSI DATASET SPECIFICATIONS.

F **(No. bands)**	**Wavelengths (*nm*)**	***X*** **(height)**	***Y*** **(width)**	**No. of classes**	**Spatial resolution (*m*)**
220	400-2500	145	145	16	20

where $i \in [1, L]$ and n_i corresponds to the amount of bands in subsegment $\mathbf{D}_i$. Then, $\mathbf{D}_1 \approx \bar{\mathbf{D}}_1$ and The desired amount of features to be retrieved from $\mathbf{D}_i$ is delineated by t_i. In a similar manner, $\mathbf{D}_2 \approx \bar{\mathbf{D}}_2$ and so forth, and the features from $\mathbf{D}_2$, ..., $\mathbf{D}_L$ can be retrieved. The final projection matrix, $\mathbf{D}^*$, which serves as the basis for classification, is formed by assembling the projection matrix of each $\mathbf{D}_i$.

III. Experimental Result

A. Dataset Description

Several studies are conducted on the Indian Pines dataset to assess the efficiency of various approaches. By dint of Grupo De Inteligencia (GIC), this dataset is retrieved from the Purdue University Research Repository (PURR) [13]. Across the Indian Pines agricultural test site near Indiana, USA, NASA's AVIRIS obtained the dataset. Table I highlights the comprehensive details of this dataset.

B. Experimental Setting

The tested HSI dataset has 21025 pixels, each class of ground truth with 16 labels, and a sizeable portion of 10776 unlabeled pixels.Following the removal of the unclassified pixels, the dataset, consisting of 10249 pixels categorized into 16 groups, has been utilized for the tests. As depicted in Figure 2, the dataset has been divided into a continual collection of bands. based on the band-to-band correlation matrix. The correlation ranges from -1.0 to 1.0, and 0.0 implies the absence of correlation. Considering the potential segments as exhibited in Figure 2,With 3, 4, or 5 segments of multiple band-subgroups in each application, SPCA and

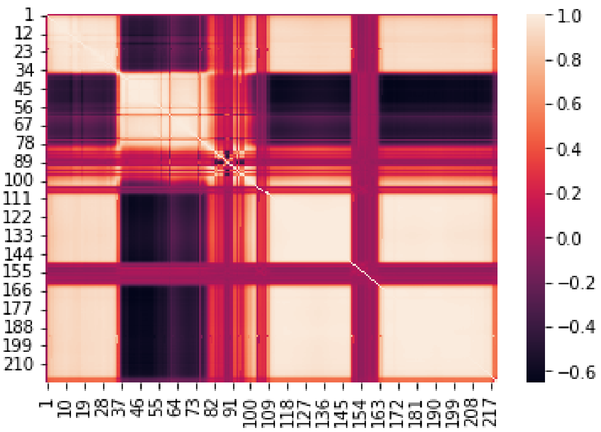

Fig. 2. Graphical Visualization of the band-to-band correlation matrix of the HSI dataset of Indian Pines.

SSPCA are referred to as SPCA3, SPCA4, and SPCA5, and SSPCA3, SSPCA4, and SSPCA5, respectively. In the Table II, we convey the specific segmentation figures.

The whole dataset has been split into three segments for each of them, as indicated by the acronyms SPCA3 and SSPCA3. Similarly, the dataset has been partitioned into 4 and 5 segments for SPCA4 and SSPCA4, as well as SPCA5 and SSPCA5. The diagonal of the correlation matrix has a mean correlation of $\approx$0.50 and it has been elected to segregate the adjoining segments. [10].

The Scikit-learn Python library's support vector machine (SVM) leveraging an RBF kernel is employed during classification in order to calculate the overall classification accuracy along with some other metrics for performance. To quantitatively evaluate the approaches, overall classification accuracy is typically considered as an objective parameter [14]. Per-pixel classification in SVM is used to classify each pixel individually. The best *C* and *gamma* values for efficient SVM model training and testing are found via 10-fold cross-validation. For improved classification performance, the training and testing samples are normalized using the *MinMaxScaler* from Scikit-learn python library. The feature-reduced dataset is then divided into training and testing datasets with a split of 80% for training and 20% for testing.

The crucial challenge with feature extraction methods resides figuring out how many Principal Components (PCs) to use for the best classification results. To determine the most ideal number of PCs for PCA, SPCA, Sparse-PCA, and SSPCA, we have conducted substantial experiments. The optimum amount of PCs are chosen for the SPCA and SSPCA segments as stated in Table II based on the conducted tests delineated in Table III. In accordance with the total classification accuracy for each one of the tested approaches, all of the tests are concluded and scored.

The number of PCs for techniques which do not employ segmentation like PCA, Sparse-PCA are selected by enumerating through all possible number of bands starting from 1 PC to the highest number of PCs by increasing it gradually [15].

TABLE II
DETAILED BAND SEGMENTATION INFORMATION FOR SPCA AND SSPCA ON THE INDIAN PINES DATASET.

Segment	Factors	Number of Segments		
		3 (SPCA3 & SSPCA3)	4 (SPCA4 & SSPCA4)	5 (SPCA5 & SSPCA5)
1	Band range No. of PCs Average correlation in diagonal	1-36 4 0.87312	1-36 4 0.87312	1-36 2 0.87312
2	Band range No. of PCs Average correlation in diagonal	37–102 4 0.54622	37–79 6 0.89601	37–79 6 0.89601
3	Band range No. of PCs Average correlation in diagonal	103–220 5 0.69487	80–102 4 0.43316	80–102 3 0.43316
4	Band range No. of PCs Average correlation in diagonal	N/A	103–220 5 0.69487	103–162 3 0.51454
5	Band range No. of PCs Average correlation in diagonal	N/A	N/A	163–220 5 0.93004

TABLE III
DETAIL OF THE CONDUCTED EXPERIMENTS TO SELECT THE PCS.

Method	Rang of PCs	No. of experiments
PCA	1-220	220
Sparse-PCA	1-220	220
SSPCA3 & SPCA3	1-16, 1-13, 1-8	1644
SSPCA4 & SPCA4	1-9, 1-7, 1-8, 1-7	3528
SSPCA5 & SPCA5	2-6, 2-6, 2-6, 2-7, 2-6	3750
Total		9362

C. Classification Performance

The training set of 3074 pixels, and the testing set of 7175 pixels, have been split from the whole dataset comprising of 10249 pixels. Samples from each of the 16 classes are included in both the training and testing sets while splitting. Due to the fact that certain classes have a significant number of pixels while others have a relatively minuscule number, the dataset has a problem with class imbalance. In Table IV, we present the classification results with regard to of the performance measures. Each approach under investigation uses the exact number of PCs that results in the optimum overall accuracy.

As demonstrated in Table IV, the proposed SSPCA3 offers the highest level of classification performance among all the approaches that have been looked into when it comes to overall accuracy, precision, recall, and scores of kappa. furthermore, the second and third best performers in that assessment, respectively, are the the proposed SSPCA4 and SSPCA5. SSPCA3, SSPCA4, and SSPCA5 require only 13, 19, and 19 extracted features, respectively, to produce the best performances. The other feature extraction methods without segmentation i.e., PCA, and Sparse-PCA requires a larger number of PCs to produce optimal classification performance.

TABLE IV
CLASSIFICATION RESULTS USING THE PROPOSED SSPCA AND THE EXISTING METHODS.

Method	No. of PCs	SVM (kernel, *C*, *gamma*)	Overall accuracy × 100	Precision score × 100	Recall score × 100	Kappa score × 100
Original dataset	220	rbf, 10, 0.11	82.997	83.117	82.997	80.500
PCA	72	rbf, 10, 0.96	83.554	83.516	83.554	81.148
Sparse PCA	105	rbf, 10, 0.44	88.39	88.398	88.39	86.734
SPCA3	15	rbf, 10, 2.60	86.091	85.930	86.091	84.085
SSPCA3	13	rbf, 10, 1.93	**91.024**	**90.968**	**90.927**	**89.625**
SPCA4	12	rbf, 10, 3.03	86.774	86.843	86.774	84.858
SSPCA4	19	rbf, 10, 1.38	90.8	90.689	90.683	89.352
SPCA5	21	rbf, 10, 1.59	86.230	86.304	86.230	84.219
SSPCA5	19	rbf, 10, 1.33	89.61	89.704	89.61	88.109

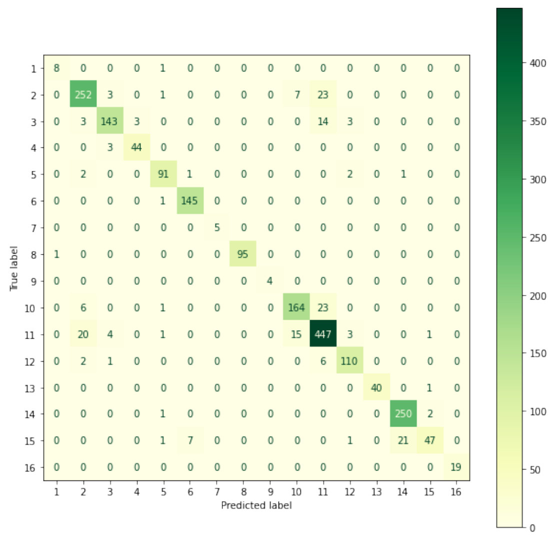

Fig. 3. Confusion matrix of the SSPCA classification.

In addition,Additionally, compared to relying on the original dataset for classification (82.997% overall accuracy) without applying any feature extraction, every feature extraction strategy yields superior result.

We finally provide the confusion matrix of the best classification performer SSPCA3 in Figure 3. The confusion matrix calculates the distribution of all expected responses and compares them to their correct classifications. Let the 2D confusion matrix be $\mathbf{C}$, where $\mathbf{C}_{ij}$ denotes the number of entries of class i that is predicted as being of class j by the classifier model. $\mathbf{C}_{ij}$ where $i = j$ denotes the the value of true positives of class i.

IV. CONCLUSION

An ameliorated as well as reliable feature extraction technique using correlation-based segmentation and Sparse-PCA, called the SSPCA has been introduced in this paper. This introduced method efficiently recuperates the local along with the global inherent attributes of the complete HSI dataset. For this approach, the data is first segmented using band-to-band correlation, and the segmented data is then fitted into the Sparse-PCA. Subsequently, The proposed SSPCA has been contrasted with a handful of renowned feature extraction technique, including PCA, SPCA, and Sparse-PCA in terms of classification performance measures. The comparison analysis demonstrates that The proposed SSPCA surpasses all of the tested feature extraction strategies while using a substantially less number of PCs. Nevertheless, the classification performances may negligibly fluctuate if the clssifier parameters, segmentation norms, etc., are impacted. Furthermore, the classification accuracy might be enhanced by providing the classifier additional training examples to choose from and by addressing the issue of class imbalance in the dataset.

REFERENCES

[1] J. A. Richards and X. Jia, "Multispectral transformations of image data," *Remote Sensing Digital Image Analysis: An Introduction*, pp. 137–163, 2006.

[2] M. A. Hossain, X. Jia, and M. Pickering, "Subspace detection using a mutual information measure for hyperspectral image classification," *IEEE Geoscience and Remote Sensing Letters*, vol. 11, no. 2, pp. 424–428, 2013.

[3] M. P. Uddin, M. A. Mamun, M. I. Afjal, and M. A. Hossain, "Information-theoretic feature selection with segmentation-based folded principal component analysis (pca) for hyperspectral image classification," *International Journal of Remote Sensing*, vol. 42, no. 1, pp. 286–321, 2021.

[4] X. Jia, B.-C. Kuo, and M. M. Crawford, "Feature mining for hyperspectral image classification," *Proceedings of the IEEE*, vol. 101, no. 3, pp. 676–697, 2013.

[5] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Transactions on geoscience and remote sensing*, vol. 42, no. 8, pp. 1778–1790, 2004.

[6] F. Tsai, E.-K. Lin, and K. Yoshino, "Spectrally segmented principal component analysis of hyperspectral imagery for mapping invasive plant species," *International Journal of Remote Sensing*, vol. 28, no. 5, pp. 1023–1039, 2007.

[7] M. Fauvel, J. Chanussot, and J. A. Benediktsson, "Kernel principal component analysis for feature reduction in hyperspectrale images analysis," in *Proceedings of the 7th Nordic Signal Processing Symposium-NORSIG 2006*. IEEE, 2006, pp. 238–241.

[8] B. Liu, X. Yu, P. Zhang, A. Yu, Q. Fu, and X. Wei, "Supervised deep feature extraction for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 56, no. 4, pp. 1909–1921, 2017.

[9] Y. Shao, N. Sang, C. Gao, and L. Ma, "Spatial and class structure regularized sparse representation graph for semi-supervised hyperspectral image classification," *Pattern Recognition*, vol. 81, pp. 81–94, 2018.

[10] X. Jia and J. A. Richards, "Segmented principal components transformation for efficient hyperspectral remote-sensing image display and classification," *IEEE transactions on Geoscience and Remote Sensing*, vol. 37, no. 1, pp. 538–542, 1999.

[11] B. Yousefi, S. Sojasi, C. I. Castanedo, G. Beaudoin, F. Huot, X. P. Maldague, M. Chamberland, and E. Lalonde, "Mineral identification in hyperspectral imaging using sparse-pca," in *Thermosense: thermal infrared applications XXXVIII*, vol. 9861. SPIE, 2016, pp. 312–322.

[12] I. M. Johnstone and A. Y. Lu, "Sparse principal components analysis," *arXiv preprint arXiv:0901.4392*, 2009.

[13] M. F. Baumgardner, L. L. Biehl, and D. A. Landgrebe, "220 band aviris hyperspectral image data set: June 12, 1992 indian pine test site 3," *Purdue University Research Repository*, vol. 10, p. R7RX991C, 2015.

[14] M. P. Uddin, M. A. Mamun, and M. A. Hossain, "Effective feature extraction through segmentation-based folded-pca for hyperspectral image classification," *International Journal of Remote Sensing*, vol. 40, no. 18, pp. 7190–7220, 2019.

[15] L. Cao, K. S. Chua, W. Chong, H. Lee, and Q. Gu, "A comparison of pca, kpca and ica for dimensionality reduction in support vector machine," *Neurocomputing*, vol. 55, no. 1-2, pp. 321–336, 2003.